```python
# This Python 3 environment comes with many helpful analytics
libraries installed
# It is defined by the kaggle/python Docker image:
https://github.com/kaggle/docker-python
# For example, here's several helpful packages to load

import numpy as np # linear algebra
import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv)

# Input data files are available in the read-only "../input/"
directory
# For example, running this (by clicking run or pressing Shift+Enter)
will list all files under the input directory

import os
for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))

# You can write up to 20GB to the current directory (/kaggle/working/)
that gets preserved as output when you create a version using "Save &
Run All"
# You can also write temporary files to /kaggle/temp/, but they won't
be saved outside of the current session
```

```
/kaggle/input/pokemon-images-and-types/pokemon.csv
/kaggle/input/pokemon-images-and-types/images/throh.png
/kaggle/input/pokemon-images-and-types/images/samurott.png
/kaggle/input/pokemon-images-and-types/images/venomoth.png
/kaggle/input/pokemon-images-and-types/images/latios.png
/kaggle/input/pokemon-images-and-types/images/shinx.png
/kaggle/input/pokemon-images-and-types/images/durant.png
/kaggle/input/pokemon-images-and-types/images/primarina.png
/kaggle/input/pokemon-images-and-types/images/geodude.png
/kaggle/input/pokemon-images-and-types/images/popplio.png
/kaggle/input/pokemon-images-and-types/images/wingull.png
/kaggle/input/pokemon-images-and-types/images/miltank.png
/kaggle/input/pokemon-images-and-types/images/castform.png
/kaggle/input/pokemon-images-and-types/images/squirtle.png
/kaggle/input/pokemon-images-and-types/images/shaymin-land.png
/kaggle/input/pokemon-images-and-types/images/raikou.png
/kaggle/input/pokemon-images-and-types/images/guzzlord.png
/kaggle/input/pokemon-images-and-types/images/tapu-bulu.png
/kaggle/input/pokemon-images-and-types/images/mawile.png
/kaggle/input/pokemon-images-and-types/images/gallade.png
/kaggle/input/pokemon-images-and-types/images/patrat.png
/kaggle/input/pokemon-images-and-types/images/froslass.png
/kaggle/input/pokemon-images-and-types/images/nidoking.png
/kaggle/input/pokemon-images-and-types/images/magnemite.png
/kaggle/input/pokemon-images-and-types/images/brionne.png
```

```
/kaggle/input/pokemon-images-and-types/images/cloyster.png
/kaggle/input/pokemon-images-and-types/images/spinarak.png
/kaggle/input/pokemon-images-and-types/images/meloetta-aria.png
/kaggle/input/pokemon-images-and-types/images/exploud.png
/kaggle/input/pokemon-images-and-types/images/vanillish.png
/kaggle/input/pokemon-images-and-types/images/scrafty.png
/kaggle/input/pokemon-images-and-types/images/azelf.png
/kaggle/input/pokemon-images-and-types/images/mew.png
/kaggle/input/pokemon-images-and-types/images/toucannon.png
/kaggle/input/pokemon-images-and-types/images/dialga.png
/kaggle/input/pokemon-images-and-types/images/cryogonal.png
/kaggle/input/pokemon-images-and-types/images/venonat.png
/kaggle/input/pokemon-images-and-types/images/scatterbug.png
/kaggle/input/pokemon-images-and-types/images/regigigas.png
/kaggle/input/pokemon-images-and-types/images/ferroseed.png
/kaggle/input/pokemon-images-and-types/images/butterfree.png
/kaggle/input/pokemon-images-and-types/images/porygon.png
/kaggle/input/pokemon-images-and-types/images/craboninable.png
/kaggle/input/pokemon-images-and-types/images/tepig.png
/kaggle/input/pokemon-images-and-types/images/dragonite.png
/kaggle/input/pokemon-images-and-types/images/paras.png
/kaggle/input/pokemon-images-and-types/images/scolipede.png
/kaggle/input/pokemon-images-and-types/images/smeargle.png
/kaggle/input/pokemon-images-and-types/images/pichu.png
/kaggle/input/pokemon-images-and-types/images/tropius.png
/kaggle/input/pokemon-images-and-types/images/typhlosion.png
/kaggle/input/pokemon-images-and-types/images/dartrix.png
/kaggle/input/pokemon-images-and-types/images/masquerain.png
/kaggle/input/pokemon-images-and-types/images/kecleon.png
/kaggle/input/pokemon-images-and-types/images/breloom.png
/kaggle/input/pokemon-images-and-types/images/inkay.png
/kaggle/input/pokemon-images-and-types/images/krabby.png
/kaggle/input/pokemon-images-and-types/images/swalot.png
/kaggle/input/pokemon-images-and-types/images/liepard.png
/kaggle/input/pokemon-images-and-types/images/cherrim.png
/kaggle/input/pokemon-images-and-types/images/taillow.png
/kaggle/input/pokemon-images-and-types/images/torracat.png
/kaggle/input/pokemon-images-and-types/images/infernape.png
/kaggle/input/pokemon-images-and-types/images/salazzle.png
/kaggle/input/pokemon-images-and-types/images/raichu.png
/kaggle/input/pokemon-images-and-types/images/hypno.png
/kaggle/input/pokemon-images-and-types/images/charmander.png
/kaggle/input/pokemon-images-and-types/images/cleffa.png
/kaggle/input/pokemon-images-and-types/images/slowking.png
/kaggle/input/pokemon-images-and-types/images/quagsire.png
/kaggle/input/pokemon-images-and-types/images/furfrou.png
/kaggle/input/pokemon-images-and-types/images/snubbull.png
/kaggle/input/pokemon-images-and-types/images/azurill.png
/kaggle/input/pokemon-images-and-types/images/kommo-o.png
```

```
/kaggle/input/pokemon-images-and-types/images/barboach.png
/kaggle/input/pokemon-images-and-types/images/kakuna.png
/kaggle/input/pokemon-images-and-types/images/mankey.png
/kaggle/input/pokemon-images-and-types/images/zebstrika.png
/kaggle/input/pokemon-images-and-types/images/dewpider.png
/kaggle/input/pokemon-images-and-types/images/aggron.png
/kaggle/input/pokemon-images-and-types/images/bronzor.png
/kaggle/input/pokemon-images-and-types/images/onix.png
/kaggle/input/pokemon-images-and-types/images/horsea.png
/kaggle/input/pokemon-images-and-types/images/vileplume.png
/kaggle/input/pokemon-images-and-types/images/emboar.png
/kaggle/input/pokemon-images-and-types/images/kricketot.png
/kaggle/input/pokemon-images-and-types/images/metagross.png
/kaggle/input/pokemon-images-and-types/images/victini.png
/kaggle/input/pokemon-images-and-types/images/ursaring.png
/kaggle/input/pokemon-images-and-types/images/garchomp.png
/kaggle/input/pokemon-images-and-types/images/sudowoodo.png
/kaggle/input/pokemon-images-and-types/images/hoppip.png
/kaggle/input/pokemon-images-and-types/images/tyranitar.png
/kaggle/input/pokemon-images-and-types/images/lotad.png
/kaggle/input/pokemon-images-and-types/images/klinklang.png
/kaggle/input/pokemon-images-and-types/images/dewott.png
/kaggle/input/pokemon-images-and-types/images/pangoro.png
/kaggle/input/pokemon-images-and-types/images/carvanha.png
/kaggle/input/pokemon-images-and-types/images/skitty.png
/kaggle/input/pokemon-images-and-types/images/tympole.png
/kaggle/input/pokemon-images-and-types/images/axew.png
/kaggle/input/pokemon-images-and-types/images/drifblim.png
/kaggle/input/pokemon-images-and-types/images/skuntank.png
/kaggle/input/pokemon-images-and-types/images/noctowl.png
/kaggle/input/pokemon-images-and-types/images/dugtrio.png
/kaggle/input/pokemon-images-and-types/images/articuno.png
/kaggle/input/pokemon-images-and-types/images/sentret.png
/kaggle/input/pokemon-images-and-types/images/loudred.png
/kaggle/input/pokemon-images-and-types/images/garbodor.png
/kaggle/input/pokemon-images-and-types/images/drampa.png
/kaggle/input/pokemon-images-and-types/images/lugia.png
/kaggle/input/pokemon-images-and-types/images/nidoran-f.png
/kaggle/input/pokemon-images-and-types/images/magcargo.png
/kaggle/input/pokemon-images-and-types/images/chimchar.png
/kaggle/input/pokemon-images-and-types/images/drifloon.png
/kaggle/input/pokemon-images-and-types/images/marill.png
/kaggle/input/pokemon-images-and-types/images/croagunk.png
/kaggle/input/pokemon-images-and-types/images/doduo.png
/kaggle/input/pokemon-images-and-types/images/dusclops.png
/kaggle/input/pokemon-images-and-types/images/glameow.png
/kaggle/input/pokemon-images-and-types/images/crobat.png
/kaggle/input/pokemon-images-and-types/images/grumpig.png
/kaggle/input/pokemon-images-and-types/images/abomasnow.png
```

```
/kaggle/input/pokemon-images-and-types/images/simisage.png
/kaggle/input/pokemon-images-and-types/images/frillish.png
/kaggle/input/pokemon-images-and-types/images/smoochum.png
/kaggle/input/pokemon-images-and-types/images/oranguru.png
/kaggle/input/pokemon-images-and-types/images/eelektrik.png
/kaggle/input/pokemon-images-and-types/images/dustox.png
/kaggle/input/pokemon-images-and-types/images/meltan.png
/kaggle/input/pokemon-images-and-types/images/ivysaur.png
/kaggle/input/pokemon-images-and-types/images/dratini.png
/kaggle/input/pokemon-images-and-types/images/bulbasaur.png
/kaggle/input/pokemon-images-and-types/images/noibat.png
/kaggle/input/pokemon-images-and-types/images/cubchoo.png
/kaggle/input/pokemon-images-and-types/images/gorebyss.png
/kaggle/input/pokemon-images-and-types/images/hitmonchan.png
/kaggle/input/pokemon-images-and-types/images/glaceon.png
/kaggle/input/pokemon-images-and-types/images/deino.png
/kaggle/input/pokemon-images-and-types/images/aurorus.png
/kaggle/input/pokemon-images-and-types/images/wynaut.png
/kaggle/input/pokemon-images-and-types/images/ninetales.png
/kaggle/input/pokemon-images-and-types/images/wishiwashi-solo.png
/kaggle/input/pokemon-images-and-types/images/pelipper.png
/kaggle/input/pokemon-images-and-types/images/petilil.png
/kaggle/input/pokemon-images-and-types/images/conkeldurr.png
/kaggle/input/pokemon-images-and-types/images/poliwrath.png
/kaggle/input/pokemon-images-and-types/images/passimian.png
/kaggle/input/pokemon-images-and-types/images/kartana.png
/kaggle/input/pokemon-images-and-types/images/whiscash.png
/kaggle/input/pokemon-images-and-types/images/flabebe.png
/kaggle/input/pokemon-images-and-types/images/seedot.png
/kaggle/input/pokemon-images-and-types/images/duosion.png
/kaggle/input/pokemon-images-and-types/images/darmanitan-standard.png
/kaggle/input/pokemon-images-and-types/images/volcarona.png
/kaggle/input/pokemon-images-and-types/images/wartortle.png
/kaggle/input/pokemon-images-and-types/images/dedenne.png
/kaggle/input/pokemon-images-and-types/images/pumpkaboo-average.png
/kaggle/input/pokemon-images-and-types/images/bellossom.png
/kaggle/input/pokemon-images-and-types/images/carbink.png
/kaggle/input/pokemon-images-and-types/images/ariados.png
/kaggle/input/pokemon-images-and-types/images/ferrothorn.png
/kaggle/input/pokemon-images-and-types/images/klink.png
/kaggle/input/pokemon-images-and-types/images/venusaur.png
/kaggle/input/pokemon-images-and-types/images/drilbur.png
/kaggle/input/pokemon-images-and-types/images/aegislash-blade.png
/kaggle/input/pokemon-images-and-types/images/karrablast.png
/kaggle/input/pokemon-images-and-types/images/golisopod.png
/kaggle/input/pokemon-images-and-types/images/sunkern.png
/kaggle/input/pokemon-images-and-types/images/golbat.png
/kaggle/input/pokemon-images-and-types/images/mudbray.png
/kaggle/input/pokemon-images-and-types/images/cranidos.png
```

```
/kaggle/input/pokemon-images-and-types/images/kangaskhan.png
/kaggle/input/pokemon-images-and-types/images/lopunny.png
/kaggle/input/pokemon-images-and-types/images/metang.png
/kaggle/input/pokemon-images-and-types/images/unfezant.png
/kaggle/input/pokemon-images-and-types/images/pawniard.png
/kaggle/input/pokemon-images-and-types/images/torterra.png
/kaggle/input/pokemon-images-and-types/images/glalie.png
/kaggle/input/pokemon-images-and-types/images/larvitar.png
/kaggle/input/pokemon-images-and-types/images/mesprit.png
/kaggle/input/pokemon-images-and-types/images/feraligatr.png
/kaggle/input/pokemon-images-and-types/images/giratina-altered.png
/kaggle/input/pokemon-images-and-types/images/pinsir.png
/kaggle/input/pokemon-images-and-types/images/dunsparce.png
/kaggle/input/pokemon-images-and-types/images/genesect.png
/kaggle/input/pokemon-images-and-types/images/skorupi.png
/kaggle/input/pokemon-images-and-types/images/slowbro.png
/kaggle/input/pokemon-images-and-types/images/leafeon.png
/kaggle/input/pokemon-images-and-types/images/diggersby.png
/kaggle/input/pokemon-images-and-types/images/girafarig.png
/kaggle/input/pokemon-images-and-types/images/shiftry.png
/kaggle/input/pokemon-images-and-types/images/staravia.png
/kaggle/input/pokemon-images-and-types/images/pyukumuku.png
/kaggle/input/pokemon-images-and-types/images/tornadus-incarnate.png
/kaggle/input/pokemon-images-and-types/images/barbaracle.png
/kaggle/input/pokemon-images-and-types/images/lunala.png
/kaggle/input/pokemon-images-and-types/images/carracosta.png
/kaggle/input/pokemon-images-and-types/images/voltorb.png
/kaggle/input/pokemon-images-and-types/images/jirachi.png
/kaggle/input/pokemon-images-and-types/images/tapu-fini.png
/kaggle/input/pokemon-images-and-types/images/mr-mime.png
/kaggle/input/pokemon-images-and-types/images/clawitzer.png
/kaggle/input/pokemon-images-and-types/images/carnivine.png
/kaggle/input/pokemon-images-and-types/images/ponyta.png
/kaggle/input/pokemon-images-and-types/images/doublade.png
/kaggle/input/pokemon-images-and-types/images/finneon.png
/kaggle/input/pokemon-images-and-types/images/swampert.png
/kaggle/input/pokemon-images-and-types/images/vaporeon.png
/kaggle/input/pokemon-images-and-types/images/mewtwo.png
/kaggle/input/pokemon-images-and-types/images/arbok.png
/kaggle/input/pokemon-images-and-types/images/furret.png
/kaggle/input/pokemon-images-and-types/images/minun.png
/kaggle/input/pokemon-images-and-types/images/bibarel.png
/kaggle/input/pokemon-images-and-types/images/parasect.png
/kaggle/input/pokemon-images-and-types/images/lapras.png
/kaggle/input/pokemon-images-and-types/images/pidove.png
/kaggle/input/pokemon-images-and-types/images/zangoose.png
/kaggle/input/pokemon-images-and-types/images/croconaw.png
/kaggle/input/pokemon-images-and-types/images/quilladin.png
/kaggle/input/pokemon-images-and-types/images/escavalier.png
```

```
/kaggle/input/pokemon-images-and-types/images/gyarados.png
/kaggle/input/pokemon-images-and-types/images/basculin-red-striped.png
/kaggle/input/pokemon-images-and-types/images/sharpedo.png
/kaggle/input/pokemon-images-and-types/images/alakazam.png
/kaggle/input/pokemon-images-and-types/images/thundurus-incarnate.png
/kaggle/input/pokemon-images-and-types/images/vibrava.png
/kaggle/input/pokemon-images-and-types/images/fomantis.png
/kaggle/input/pokemon-images-and-types/images/bunnelby.png
/kaggle/input/pokemon-images-and-types/images/bastiodon.png
/kaggle/input/pokemon-images-and-types/images/nidorina.png
/kaggle/input/pokemon-images-and-types/images/fearow.png
/kaggle/input/pokemon-images-and-types/images/silvally.png
/kaggle/input/pokemon-images-and-types/images/medicham.png
/kaggle/input/pokemon-images-and-types/images/shuppet.png
/kaggle/input/pokemon-images-and-types/images/pikipek.png
/kaggle/input/pokemon-images-and-types/images/gulpin.png
/kaggle/input/pokemon-images-and-types/images/groudon.png
/kaggle/input/pokemon-images-and-types/images/cubone.png
/kaggle/input/pokemon-images-and-types/images/gothita.png
/kaggle/input/pokemon-images-and-types/images/zigzagoon.png
/kaggle/input/pokemon-images-and-types/images/sceptile.png
/kaggle/input/pokemon-images-and-types/images/zekrom.png
/kaggle/input/pokemon-images-and-types/images/volbeat.png
/kaggle/input/pokemon-images-and-types/images/oddish.png
/kaggle/input/pokemon-images-and-types/images/beldum.png
/kaggle/input/pokemon-images-and-types/images/kingler.png
/kaggle/input/pokemon-images-and-types/images/hitmonlee.png
/kaggle/input/pokemon-images-and-types/images/togepi.png
/kaggle/input/pokemon-images-and-types/images/spearow.png
/kaggle/input/pokemon-images-and-types/images/armaldo.png
/kaggle/input/pokemon-images-and-types/images/latias.png
/kaggle/input/pokemon-images-and-types/images/tyrogue.png
/kaggle/input/pokemon-images-and-types/images/avalugg.png
/kaggle/input/pokemon-images-and-types/images/banette.png
/kaggle/input/pokemon-images-and-types/images/illumise.png
/kaggle/input/pokemon-images-and-types/images/lairon.png
/kaggle/input/pokemon-images-and-types/images/crustle.png
/kaggle/input/pokemon-images-and-types/images/sandslash.png
/kaggle/input/pokemon-images-and-types/images/diglett.png
/kaggle/input/pokemon-images-and-types/images/anorith.png
/kaggle/input/pokemon-images-and-types/images/vigoroth.png
/kaggle/input/pokemon-images-and-types/images/mienshao.png
/kaggle/input/pokemon-images-and-types/images/gumshoos.png
/kaggle/input/pokemon-images-and-types/images/jolteon.png
/kaggle/input/pokemon-images-and-types/images/electabuzz.png
/kaggle/input/pokemon-images-and-types/images/skiddo.png
/kaggle/input/pokemon-images-and-types/images/combusken.png
/kaggle/input/pokemon-images-and-types/images/haunter.png
/kaggle/input/pokemon-images-and-types/images/luxray.png
```

```
/kaggle/input/pokemon-images-and-types/images/whimsicott.png
/kaggle/input/pokemon-images-and-types/images/dodrio.png
/kaggle/input/pokemon-images-and-types/images/porygon-z.png
/kaggle/input/pokemon-images-and-types/images/dwebble.png
/kaggle/input/pokemon-images-and-types/images/tapu-koko.png
/kaggle/input/pokemon-images-and-types/images/salandit.png
/kaggle/input/pokemon-images-and-types/images/lileep.png
/kaggle/input/pokemon-images-and-types/images/azumarill.png
/kaggle/input/pokemon-images-and-types/images/serperior.png
/kaggle/input/pokemon-images-and-types/images/zorua.png
/kaggle/input/pokemon-images-and-types/images/golurk.png
/kaggle/input/pokemon-images-and-types/images/spinda.png
/kaggle/input/pokemon-images-and-types/images/panpour.png
/kaggle/input/pokemon-images-and-types/images/pidgey.png
/kaggle/input/pokemon-images-and-types/images/simipour.png
/kaggle/input/pokemon-images-and-types/images/clamperl.png
/kaggle/input/pokemon-images-and-types/images/goldeen.png
/kaggle/input/pokemon-images-and-types/images/plusle.png
/kaggle/input/pokemon-images-and-types/images/cresselia.png
/kaggle/input/pokemon-images-and-types/images/xatu.png
/kaggle/input/pokemon-images-and-types/images/sandygast.png
/kaggle/input/pokemon-images-and-types/images/weavile.png
/kaggle/input/pokemon-images-and-types/images/heracross.png
/kaggle/input/pokemon-images-and-types/images/houndour.png
/kaggle/input/pokemon-images-and-types/images/bisharp.png
/kaggle/input/pokemon-images-and-types/images/herdier.png
/kaggle/input/pokemon-images-and-types/images/granbull.png
/kaggle/input/pokemon-images-and-types/images/pignite.png
/kaggle/input/pokemon-images-and-types/images/delcatty.png
/kaggle/input/pokemon-images-and-types/images/roselia.png
/kaggle/input/pokemon-images-and-types/images/gastly.png
/kaggle/input/pokemon-images-and-types/images/cacturne.png
/kaggle/input/pokemon-images-and-types/images/sylveon.png
/kaggle/input/pokemon-images-and-types/images/goomy.png
/kaggle/input/pokemon-images-and-types/images/bagon.png
/kaggle/input/pokemon-images-and-types/images/golett.png
/kaggle/input/pokemon-images-and-types/images/phantump.png
/kaggle/input/pokemon-images-and-types/images/floette.png
/kaggle/input/pokemon-images-and-types/images/togetic.png
/kaggle/input/pokemon-images-and-types/images/bonsly.png
/kaggle/input/pokemon-images-and-types/images/eevee.png
/kaggle/input/pokemon-images-and-types/images/numel.png
/kaggle/input/pokemon-images-and-types/images/floatzel.png
/kaggle/input/pokemon-images-and-types/images/eelektross.png
/kaggle/input/pokemon-images-and-types/images/shroomish.png
/kaggle/input/pokemon-images-and-types/images/misdreavus.png
/kaggle/input/pokemon-images-and-types/images/natu.png
/kaggle/input/pokemon-images-and-types/images/lillipup.png
/kaggle/input/pokemon-images-and-types/images/lickilicky.png
```

```
/kaggle/input/pokemon-images-and-types/images/sunflora.png
/kaggle/input/pokemon-images-and-types/images/vullaby.png
/kaggle/input/pokemon-images-and-types/images/swinub.png
/kaggle/input/pokemon-images-and-types/images/larvesta.png
/kaggle/input/pokemon-images-and-types/images/swoobat.png
/kaggle/input/pokemon-images-and-types/images/kingdra.png
/kaggle/input/pokemon-images-and-types/images/druddigon.png
/kaggle/input/pokemon-images-and-types/images/sableye.png
/kaggle/input/pokemon-images-and-types/images/litwick.png
/kaggle/input/pokemon-images-and-types/images/gardevoir.png
/kaggle/input/pokemon-images-and-types/images/cinccino.png
/kaggle/input/pokemon-images-and-types/images/ekans.png
/kaggle/input/pokemon-images-and-types/images/blastoise.png
/kaggle/input/pokemon-images-and-types/images/keldeo-ordinary.png
/kaggle/input/pokemon-images-and-types/images/makuhita.png
/kaggle/input/pokemon-images-and-types/images/luvdisc.png
/kaggle/input/pokemon-images-and-types/images/pupitar.png
/kaggle/input/pokemon-images-and-types/images/melmetal.png
/kaggle/input/pokemon-images-and-types/images/magmar.png
/kaggle/input/pokemon-images-and-types/images/surskit.png
/kaggle/input/pokemon-images-and-types/images/empoleon.png
/kaggle/input/pokemon-images-and-types/images/mudkip.png
/kaggle/input/pokemon-images-and-types/images/yamask.png
/kaggle/input/pokemon-images-and-types/images/delibird.png
/kaggle/input/pokemon-images-and-types/images/seadra.png
/kaggle/input/pokemon-images-and-types/images/xurkitree.png
/kaggle/input/pokemon-images-and-types/images/litten.png
/kaggle/input/pokemon-images-and-types/images/kirlia.png
/kaggle/input/pokemon-images-and-types/images/chingling.png
/kaggle/input/pokemon-images-and-types/images/excadrill.png
/kaggle/input/pokemon-images-and-types/images/archeops.png
/kaggle/input/pokemon-images-and-types/images/landorus-incarnate.png
/kaggle/input/pokemon-images-and-types/images/arceus.png
/kaggle/input/pokemon-images-and-types/images/blissey.png
/kaggle/input/pokemon-images-and-types/images/sandshrew.png
/kaggle/input/pokemon-images-and-types/images/sawk.png
/kaggle/input/pokemon-images-and-types/images/whismur.png
/kaggle/input/pokemon-images-and-types/images/bronzong.png
/kaggle/input/pokemon-images-and-types/images/wigglytuff.png
/kaggle/input/pokemon-images-and-types/images/accelgor.png
/kaggle/input/pokemon-images-and-types/images/honchkrow.png
/kaggle/input/pokemon-images-and-types/images/rowlet.png
/kaggle/input/pokemon-images-and-types/images/tirtouga.png
/kaggle/input/pokemon-images-and-types/images/whirlipede.png
/kaggle/input/pokemon-images-and-types/images/ditto.png
/kaggle/input/pokemon-images-and-types/images/budew.png
/kaggle/input/pokemon-images-and-types/images/florges.png
/kaggle/input/pokemon-images-and-types/images/diancie.png
/kaggle/input/pokemon-images-and-types/images/grotle.png
```

```
/kaggle/input/pokemon-images-and-types/images/murkrow.png
/kaggle/input/pokemon-images-and-types/images/venipede.png
/kaggle/input/pokemon-images-and-types/images/krookodile.png
/kaggle/input/pokemon-images-and-types/images/machop.png
/kaggle/input/pokemon-images-and-types/images/buneary.png
/kaggle/input/pokemon-images-and-types/images/kyurem.png
/kaggle/input/pokemon-images-and-types/images/slowpoke.png
/kaggle/input/pokemon-images-and-types/images/deoxys-normal.png
/kaggle/input/pokemon-images-and-types/images/manaphy.png
/kaggle/input/pokemon-images-and-types/images/nuzleaf.png
/kaggle/input/pokemon-images-and-types/images/tyrunt.png
/kaggle/input/pokemon-images-and-types/images/solrock.png
/kaggle/input/pokemon-images-and-types/images/chimecho.png
/kaggle/input/pokemon-images-and-types/images/gothorita.png
/kaggle/input/pokemon-images-and-types/images/kadabra.png
/kaggle/input/pokemon-images-and-types/images/happiny.png
/kaggle/input/pokemon-images-and-types/images/weepinbell.png
/kaggle/input/pokemon-images-and-types/images/seismitoad.png
/kaggle/input/pokemon-images-and-types/images/clauncher.png
/kaggle/input/pokemon-images-and-types/images/pineco.png
/kaggle/input/pokemon-images-and-types/images/camerupt.png
/kaggle/input/pokemon-images-and-types/images/dhelmise.png
/kaggle/input/pokemon-images-and-types/images/forretress.png
/kaggle/input/pokemon-images-and-types/images/fennekin.png
/kaggle/input/pokemon-images-and-types/images/corphish.png
/kaggle/input/pokemon-images-and-types/images/swellow.png
/kaggle/input/pokemon-images-and-types/images/pancham.png
/kaggle/input/pokemon-images-and-types/images/vanilluxe.png
/kaggle/input/pokemon-images-and-types/images/litleo.png
/kaggle/input/pokemon-images-and-types/images/starmie.png
/kaggle/input/pokemon-images-and-types/images/boldore.png
/kaggle/input/pokemon-images-and-types/images/flygon.png
/kaggle/input/pokemon-images-and-types/images/rhydon.png
/kaggle/input/pokemon-images-and-types/images/jumpluff.png
/kaggle/input/pokemon-images-and-types/images/wobbuffet.png
/kaggle/input/pokemon-images-and-types/images/snorlax.png
/kaggle/input/pokemon-images-and-types/images/blacephalon.png
/kaggle/input/pokemon-images-and-types/images/stunky.png
/kaggle/input/pokemon-images-and-types/images/maractus.png
/kaggle/input/pokemon-images-and-types/images/skarmory.png
/kaggle/input/pokemon-images-and-types/images/zweilous.png
/kaggle/input/pokemon-images-and-types/images/sigilyph.png
/kaggle/input/pokemon-images-and-types/images/rayquaza.png
/kaggle/input/pokemon-images-and-types/images/mienfoo.png
/kaggle/input/pokemon-images-and-types/images/rhyperior.png
/kaggle/input/pokemon-images-and-types/images/chatot.png
/kaggle/input/pokemon-images-and-types/images/lumineon.png
/kaggle/input/pokemon-images-and-types/images/binacle.png
/kaggle/input/pokemon-images-and-types/images/lunatone.png
```

```
/kaggle/input/pokemon-images-and-types/images/magnezone.png
/kaggle/input/pokemon-images-and-types/images/toxicroak.png
/kaggle/input/pokemon-images-and-types/images/absol.png
/kaggle/input/pokemon-images-and-types/images/zapdos.png
/kaggle/input/pokemon-images-and-types/images/munchlax.png
/kaggle/input/pokemon-images-and-types/images/gastrodon.png
/kaggle/input/pokemon-images-and-types/images/omanyte.png
/kaggle/input/pokemon-images-and-types/images/phione.png
/kaggle/input/pokemon-images-and-types/images/spiritomb.png
/kaggle/input/pokemon-images-and-types/images/quilava.png
/kaggle/input/pokemon-images-and-types/images/dragonair.png
/kaggle/input/pokemon-images-and-types/images/reuniclus.png
/kaggle/input/pokemon-images-and-types/images/seel.png
/kaggle/input/pokemon-images-and-types/images/archen.png
/kaggle/input/pokemon-images-and-types/images/swanna.png
/kaggle/input/pokemon-images-and-types/images/chinchou.png
/kaggle/input/pokemon-images-and-types/images/watchog.png
/kaggle/input/pokemon-images-and-types/images/gengar.png
/kaggle/input/pokemon-images-and-types/images/emolga.png
/kaggle/input/pokemon-images-and-types/images/comfey.png
/kaggle/input/pokemon-images-and-types/images/machamp.png
/kaggle/input/pokemon-images-and-types/images/ho-oh.png
/kaggle/input/pokemon-images-and-types/images/piloswine.png
/kaggle/input/pokemon-images-and-types/images/machoke.png
/kaggle/input/pokemon-images-and-types/images/munna.png
/kaggle/input/pokemon-images-and-types/images/yungoos.png
/kaggle/input/pokemon-images-and-types/images/noivern.png
/kaggle/input/pokemon-images-and-types/images/weezing.png
/kaggle/input/pokemon-images-and-types/images/xerneas.png
/kaggle/input/pokemon-images-and-types/images/farfetchd.png
/kaggle/input/pokemon-images-and-types/images/nidoqueen.png
/kaggle/input/pokemon-images-and-types/images/nidoran-m.png
/kaggle/input/pokemon-images-and-types/images/huntail.png
/kaggle/input/pokemon-images-and-types/images/kyogre.png
/kaggle/input/pokemon-images-and-types/images/roserade.png
/kaggle/input/pokemon-images-and-types/images/golduck.png
/kaggle/input/pokemon-images-and-types/images/skrelp.png
/kaggle/input/pokemon-images-and-types/images/seviper.png
/kaggle/input/pokemon-images-and-types/images/tapu-lele.png
/kaggle/input/pokemon-images-and-types/images/mismagius.png
/kaggle/input/pokemon-images-and-types/images/exeggcute.png
/kaggle/input/pokemon-images-and-types/images/torchic.png
/kaggle/input/pokemon-images-and-types/images/lombre.png
/kaggle/input/pokemon-images-and-types/images/aipom.png
/kaggle/input/pokemon-images-and-types/images/clefairy.png
/kaggle/input/pokemon-images-and-types/images/hydreigon.png
/kaggle/input/pokemon-images-and-types/images/raticate.png
/kaggle/input/pokemon-images-and-types/images/kabutops.png
/kaggle/input/pokemon-images-and-types/images/staryu.png
```

```
/kaggle/input/pokemon-images-and-types/images/gurdurr.png
/kaggle/input/pokemon-images-and-types/images/palossand.png
/kaggle/input/pokemon-images-and-types/images/blaziken.png
/kaggle/input/pokemon-images-and-types/images/mandibuzz.png
/kaggle/input/pokemon-images-and-types/images/corsola.png
/kaggle/input/pokemon-images-and-types/images/walrein.png
/kaggle/input/pokemon-images-and-types/images/araquanid.png
/kaggle/input/pokemon-images-and-types/images/yveltal.png
/kaggle/input/pokemon-images-and-types/images/braviary.png
/kaggle/input/pokemon-images-and-types/images/ninjask.png
/kaggle/input/pokemon-images-and-types/images/tsareena.png
/kaggle/input/pokemon-images-and-types/images/snivy.png
/kaggle/input/pokemon-images-and-types/images/unown.png
/kaggle/input/pokemon-images-and-types/images/turtwig.png
/kaggle/input/pokemon-images-and-types/images/cofagrigus.png
/kaggle/input/pokemon-images-and-types/images/kricketune.png
/kaggle/input/pokemon-images-and-types/images/zygarde-50.png
/kaggle/input/pokemon-images-and-types/images/seaking.png
/kaggle/input/pokemon-images-and-types/images/buzzwole.png
/kaggle/input/pokemon-images-and-types/images/galvantula.png
/kaggle/input/pokemon-images-and-types/images/flaaffy.png
/kaggle/input/pokemon-images-and-types/images/cosmog.png
/kaggle/input/pokemon-images-and-types/images/marshtomp.png
/kaggle/input/pokemon-images-and-types/images/purugly.png
/kaggle/input/pokemon-images-and-types/images/kabuto.png
/kaggle/input/pokemon-images-and-types/images/amoonguss.png
/kaggle/input/pokemon-images-and-types/images/totodile.png
/kaggle/input/pokemon-images-and-types/images/combee.png
/kaggle/input/pokemon-images-and-types/images/vanillite.png
/kaggle/input/pokemon-images-and-types/images/pachirisu.png
/kaggle/input/pokemon-images-and-types/images/fraxure.png
/kaggle/input/pokemon-images-and-types/images/weedle.png
/kaggle/input/pokemon-images-and-types/images/hawlucha.png
/kaggle/input/pokemon-images-and-types/images/heatmor.png
/kaggle/input/pokemon-images-and-types/images/torkoal.png
/kaggle/input/pokemon-images-and-types/images/mudsdale.png
/kaggle/input/pokemon-images-and-types/images/lucario.png
/kaggle/input/pokemon-images-and-types/images/snorunt.png
/kaggle/input/pokemon-images-and-types/images/mareanie.png
/kaggle/input/pokemon-images-and-types/images/hippowdon.png
/kaggle/input/pokemon-images-and-types/images/aromatisse.png
/kaggle/input/pokemon-images-and-types/images/probopass.png
/kaggle/input/pokemon-images-and-types/images/golem.png
/kaggle/input/pokemon-images-and-types/images/nidorino.png
/kaggle/input/pokemon-images-and-types/images/umbreon.png
/kaggle/input/pokemon-images-and-types/images/sewaddle.png
/kaggle/input/pokemon-images-and-types/images/meowstic-male.png
/kaggle/input/pokemon-images-and-types/images/charmeleon.png
/kaggle/input/pokemon-images-and-types/images/delphox.png
/kaggle/input/pokemon-images-and-types/images/helioptile.png
```

```
/kaggle/input/pokemon-images-and-types/images/tauros.png
/kaggle/input/pokemon-images-and-types/images/purrloin.png
/kaggle/input/pokemon-images-and-types/images/deerling.png
/kaggle/input/pokemon-images-and-types/images/aron.png
/kaggle/input/pokemon-images-and-types/images/stufful.png
/kaggle/input/pokemon-images-and-types/images/spheal.png
/kaggle/input/pokemon-images-and-types/images/ledian.png
/kaggle/input/pokemon-images-and-types/images/regice.png
/kaggle/input/pokemon-images-and-types/images/vivillon.png
/kaggle/input/pokemon-images-and-types/images/cyndaquil.png
/kaggle/input/pokemon-images-and-types/images/beartic.png
/kaggle/input/pokemon-images-and-types/images/growlithe.png
/kaggle/input/pokemon-images-and-types/images/swadloon.png
/kaggle/input/pokemon-images-and-types/images/volcanion.png
/kaggle/input/pokemon-images-and-types/images/lanturn.png
/kaggle/input/pokemon-images-and-types/images/nihilego.png
/kaggle/input/pokemon-images-and-types/images/trapinch.png
/kaggle/input/pokemon-images-and-types/images/solgaleo.png
/kaggle/input/pokemon-images-and-types/images/pansage.png
/kaggle/input/pokemon-images-and-types/images/yanmega.png
/kaggle/input/pokemon-images-and-types/images/meganium.png
/kaggle/input/pokemon-images-and-types/images/houndoom.png
/kaggle/input/pokemon-images-and-types/images/luxio.png
/kaggle/input/pokemon-images-and-types/images/sandile.png
/kaggle/input/pokemon-images-and-types/images/salamence.png
/kaggle/input/pokemon-images-and-types/images/mightyena.png
/kaggle/input/pokemon-images-and-types/images/alomomola.png
/kaggle/input/pokemon-images-and-types/images/electrode.png
/kaggle/input/pokemon-images-and-types/images/slurpuff.png
/kaggle/input/pokemon-images-and-types/images/beedrill.png
/kaggle/input/pokemon-images-and-types/images/politoed.png
/kaggle/input/pokemon-images-and-types/images/cutiefly.png
/kaggle/input/pokemon-images-and-types/images/vulpix.png
/kaggle/input/pokemon-images-and-types/images/poochyena.png
/kaggle/input/pokemon-images-and-types/images/trevenant.png
/kaggle/input/pokemon-images-and-types/images/rotom.png
/kaggle/input/pokemon-images-and-types/images/virizion.png
/kaggle/input/pokemon-images-and-types/images/gogoat.png
/kaggle/input/pokemon-images-and-types/images/decidueye.png
/kaggle/input/pokemon-images-and-types/images/ampharos.png
/kaggle/input/pokemon-images-and-types/images/timburr.png
/kaggle/input/pokemon-images-and-types/images/ambipom.png
/kaggle/input/pokemon-images-and-types/images/bewear.png
/kaggle/input/pokemon-images-and-types/images/oricorio-baile.png
/kaggle/input/pokemon-images-and-types/images/porygon2.png
/kaggle/input/pokemon-images-and-types/images/charjabug.png
/kaggle/input/pokemon-images-and-types/images/rattata.png
/kaggle/input/pokemon-images-and-types/images/gourgeist-average.png
/kaggle/input/pokemon-images-and-types/images/wailord.png
```

```
/kaggle/input/pokemon-images-and-types/images/caterpie.png
/kaggle/input/pokemon-images-and-types/images/bayleef.png
/kaggle/input/pokemon-images-and-types/images/slugma.png
/kaggle/input/pokemon-images-and-types/images/wailmer.png
/kaggle/input/pokemon-images-and-types/images/scizor.png
/kaggle/input/pokemon-images-and-types/images/mareep.png
/kaggle/input/pokemon-images-and-types/images/swablu.png
/kaggle/input/pokemon-images-and-types/images/rockruff.png
/kaggle/input/pokemon-images-and-types/images/vikavolt.png
/kaggle/input/pokemon-images-and-types/images/terrakion.png
/kaggle/input/pokemon-images-and-types/images/aerodactyl.png
/kaggle/input/pokemon-images-and-types/images/gliscor.png
/kaggle/input/pokemon-images-and-types/images/cottonee.png
/kaggle/input/pokemon-images-and-types/images/entei.png
/kaggle/input/pokemon-images-and-types/images/jigglypuff.png
/kaggle/input/pokemon-images-and-types/images/beheeyem.png
/kaggle/input/pokemon-images-and-types/images/magmortar.png
/kaggle/input/pokemon-images-and-types/images/crawdaunt.png
/kaggle/input/pokemon-images-and-types/images/gothitelle.png
/kaggle/input/pokemon-images-and-types/images/mantine.png
/kaggle/input/pokemon-images-and-types/images/remoraid.png
/kaggle/input/pokemon-images-and-types/images/rampardos.png
/kaggle/input/pokemon-images-and-types/images/turtonator.png
/kaggle/input/pokemon-images-and-types/images/toxapex.png
/kaggle/input/pokemon-images-and-types/images/incineroar.png
/kaggle/input/pokemon-images-and-types/images/donphan.png
/kaggle/input/pokemon-images-and-types/images/hakamo-o.png
/kaggle/input/pokemon-images-and-types/images/bidoof.png
/kaggle/input/pokemon-images-and-types/images/klang.png
/kaggle/input/pokemon-images-and-types/images/fletchinder.png
/kaggle/input/pokemon-images-and-types/images/burmy.png
/kaggle/input/pokemon-images-and-types/images/grubbin.png
/kaggle/input/pokemon-images-and-types/images/duskull.png
/kaggle/input/pokemon-images-and-types/images/skiploom.png
/kaggle/input/pokemon-images-and-types/images/tentacool.png
/kaggle/input/pokemon-images-and-types/images/honedge.png
/kaggle/input/pokemon-images-and-types/images/grovyle.png
/kaggle/input/pokemon-images-and-types/images/dewgong.png
/kaggle/input/pokemon-images-and-types/images/rapidash.png
/kaggle/input/pokemon-images-and-types/images/frogadier.png
/kaggle/input/pokemon-images-and-types/images/starly.png
/kaggle/input/pokemon-images-and-types/images/froakie.png
/kaggle/input/pokemon-images-and-types/images/cradily.png
/kaggle/input/pokemon-images-and-types/images/graveler.png
/kaggle/input/pokemon-images-and-types/images/lurantis.png
/kaggle/input/pokemon-images-and-types/images/elekid.png
/kaggle/input/pokemon-images-and-types/images/steenee.png
/kaggle/input/pokemon-images-and-types/images/psyduck.png
/kaggle/input/pokemon-images-and-types/images/cobalion.png
```

```
/kaggle/input/pokemon-images-and-types/images/mantyke.png
/kaggle/input/pokemon-images-and-types/images/darkrai.png
/kaggle/input/pokemon-images-and-types/images/victreebel.png
/kaggle/input/pokemon-images-and-types/images/wooper.png
/kaggle/input/pokemon-images-and-types/images/crabrawler.png
/kaggle/input/pokemon-images-and-types/images/stantler.png
/kaggle/input/pokemon-images-and-types/images/jangmo-o.png
/kaggle/input/pokemon-images-and-types/images/magikarp.png
/kaggle/input/pokemon-images-and-types/images/darumaka.png
/kaggle/input/pokemon-images-and-types/images/tangela.png
/kaggle/input/pokemon-images-and-types/images/koffing.png
/kaggle/input/pokemon-images-and-types/images/rufflet.png
/kaggle/input/pokemon-images-and-types/images/drowzee.png
/kaggle/input/pokemon-images-and-types/images/krokorok.png
/kaggle/input/pokemon-images-and-types/images/tynamo.png
/kaggle/input/pokemon-images-and-types/images/naganadel.png
/kaggle/input/pokemon-images-and-types/images/heliolisk.png
/kaggle/input/pokemon-images-and-types/images/espurr.png
/kaggle/input/pokemon-images-and-types/images/wormadam-plant.png
/kaggle/input/pokemon-images-and-types/images/minccino.png
/kaggle/input/pokemon-images-and-types/images/stakataka.png
/kaggle/input/pokemon-images-and-types/images/ducklett.png
/kaggle/input/pokemon-images-and-types/images/type-null.png
/kaggle/input/pokemon-images-and-types/images/palpitoad.png
/kaggle/input/pokemon-images-and-types/images/tangrowth.png
/kaggle/input/pokemon-images-and-types/images/hoothoot.png
/kaggle/input/pokemon-images-and-types/images/magby.png
/kaggle/input/pokemon-images-and-types/images/rhyhorn.png
/kaggle/input/pokemon-images-and-types/images/swirlix.png
/kaggle/input/pokemon-images-and-types/images/mime-jr.png
/kaggle/input/pokemon-images-and-types/images/grimer.png
/kaggle/input/pokemon-images-and-types/images/electrike.png
/kaggle/input/pokemon-images-and-types/images/sneasel.png
/kaggle/input/pokemon-images-and-types/images/mamoswine.png
/kaggle/input/pokemon-images-and-types/images/chikorita.png
/kaggle/input/pokemon-images-and-types/images/espeon.png
/kaggle/input/pokemon-images-and-types/images/servine.png
/kaggle/input/pokemon-images-and-types/images/zoroark.png
/kaggle/input/pokemon-images-and-types/images/cascoon.png
/kaggle/input/pokemon-images-and-types/images/poliwag.png
/kaggle/input/pokemon-images-and-types/images/klefki.png
/kaggle/input/pokemon-images-and-types/images/sealeo.png
/kaggle/input/pokemon-images-and-types/images/poipole.png
/kaggle/input/pokemon-images-and-types/images/prinplup.png
/kaggle/input/pokemon-images-and-types/images/bergmite.png
/kaggle/input/pokemon-images-and-types/images/phanpy.png
/kaggle/input/pokemon-images-and-types/images/bruxish.png
/kaggle/input/pokemon-images-and-types/images/snover.png
/kaggle/input/pokemon-images-and-types/images/sliggoo.png
```

```
/kaggle/input/pokemon-images-and-types/images/claydol.png
/kaggle/input/pokemon-images-and-types/images/pidgeotto.png
/kaggle/input/pokemon-images-and-types/images/necrozma.png
/kaggle/input/pokemon-images-and-types/images/amaura.png
/kaggle/input/pokemon-images-and-types/images/hariyama.png
/kaggle/input/pokemon-images-and-types/images/wurmple.png
/kaggle/input/pokemon-images-and-types/images/buizel.png
/kaggle/input/pokemon-images-and-types/images/milotic.png
/kaggle/input/pokemon-images-and-types/images/marshadow.png
/kaggle/input/pokemon-images-and-types/images/yanma.png
/kaggle/input/pokemon-images-and-types/images/charizard.png
/kaggle/input/pokemon-images-and-types/images/arcanine.png
/kaggle/input/pokemon-images-and-types/images/metapod.png
/kaggle/input/pokemon-images-and-types/images/bouffalant.png
/kaggle/input/pokemon-images-and-types/images/gligar.png
/kaggle/input/pokemon-images-and-types/images/cosmoem.png
/kaggle/input/pokemon-images-and-types/images/persian.png
/kaggle/input/pokemon-images-and-types/images/spewpa.png
/kaggle/input/pokemon-images-and-types/images/talonflame.png
/kaggle/input/pokemon-images-and-types/images/regirock.png
/kaggle/input/pokemon-images-and-types/images/silcoon.png
/kaggle/input/pokemon-images-and-types/images/teddiursa.png
/kaggle/input/pokemon-images-and-types/images/staraptor.png
/kaggle/input/pokemon-images-and-types/images/joltik.png
/kaggle/input/pokemon-images-and-types/images/ralts.png
/kaggle/input/pokemon-images-and-types/images/shiinotic.png
/kaggle/input/pokemon-images-and-types/images/suicune.png
/kaggle/input/pokemon-images-and-types/images/slaking.png
/kaggle/input/pokemon-images-and-types/images/beautifly.png
/kaggle/input/pokemon-images-and-types/images/altaria.png
/kaggle/input/pokemon-images-and-types/images/primeape.png
/kaggle/input/pokemon-images-and-types/images/hitmontop.png
/kaggle/input/pokemon-images-and-types/images/jellicent.png
/kaggle/input/pokemon-images-and-types/images/registeel.png
/kaggle/input/pokemon-images-and-types/images/slakoth.png
/kaggle/input/pokemon-images-and-types/images/tyrantrum.png
/kaggle/input/pokemon-images-and-types/images/mimikyu.png
/kaggle/input/pokemon-images-and-types/images/bellsprout.png
/kaggle/input/pokemon-images-and-types/images/manectric.png
/kaggle/input/pokemon-images-and-types/images/fletchling.png
/kaggle/input/pokemon-images-and-types/images/shelmet.png
/kaggle/input/pokemon-images-and-types/images/foongus.png
/kaggle/input/pokemon-images-and-types/images/celebi.png
/kaggle/input/pokemon-images-and-types/images/shelgon.png
/kaggle/input/pokemon-images-and-types/images/shellos.png
/kaggle/input/pokemon-images-and-types/images/tentacruel.png
/kaggle/input/pokemon-images-and-types/images/linoone.png
/kaggle/input/pokemon-images-and-types/images/lycanroc-midday.png
/kaggle/input/pokemon-images-and-types/images/baltoy.png
```

```
/kaggle/input/pokemon-images-and-types/images/pikachu.png
/kaggle/input/pokemon-images-and-types/images/trumbeak.png
/kaggle/input/pokemon-images-and-types/images/spoink.png
/kaggle/input/pokemon-images-and-types/images/bounsweet.png
/kaggle/input/pokemon-images-and-types/images/zeraora.png
/kaggle/input/pokemon-images-and-types/images/abra.png
/kaggle/input/pokemon-images-and-types/images/relicanth.png
/kaggle/input/pokemon-images-and-types/images/goodra.png
/kaggle/input/pokemon-images-and-types/images/meditite.png
/kaggle/input/pokemon-images-and-types/images/vespiquen.png
/kaggle/input/pokemon-images-and-types/images/dragalge.png
/kaggle/input/pokemon-images-and-types/images/flareon.png
/kaggle/input/pokemon-images-and-types/images/leavanny.png
/kaggle/input/pokemon-images-and-types/images/solosis.png
/kaggle/input/pokemon-images-and-types/images/lampent.png
/kaggle/input/pokemon-images-and-types/images/shuckle.png
/kaggle/input/pokemon-images-and-types/images/elgyem.png
/kaggle/input/pokemon-images-and-types/images/chespin.png
/kaggle/input/pokemon-images-and-types/images/pidgeot.png
/kaggle/input/pokemon-images-and-types/images/woobat.png
/kaggle/input/pokemon-images-and-types/images/palkia.png
/kaggle/input/pokemon-images-and-types/images/oshawott.png
/kaggle/input/pokemon-images-and-types/images/braixen.png
/kaggle/input/pokemon-images-and-types/images/marowak.png
/kaggle/input/pokemon-images-and-types/images/zubat.png
/kaggle/input/pokemon-images-and-types/images/treecko.png
/kaggle/input/pokemon-images-and-types/images/pheromosa.png
/kaggle/input/pokemon-images-and-types/images/scyther.png
/kaggle/input/pokemon-images-and-types/images/octillery.png
/kaggle/input/pokemon-images-and-types/images/gloom.png
/kaggle/input/pokemon-images-and-types/images/scraggy.png
/kaggle/input/pokemon-images-and-types/images/tranquill.png
/kaggle/input/pokemon-images-and-types/images/steelix.png
/kaggle/input/pokemon-images-and-types/images/lickitung.png
/kaggle/input/pokemon-images-and-types/images/cherubi.png
/kaggle/input/pokemon-images-and-types/images/komala.png
/kaggle/input/pokemon-images-and-types/images/chesnaught.png
/kaggle/input/pokemon-images-and-types/images/simisear.png
/kaggle/input/pokemon-images-and-types/images/qwilfish.png
/kaggle/input/pokemon-images-and-types/images/electivire.png
/kaggle/input/pokemon-images-and-types/images/nincada.png
/kaggle/input/pokemon-images-and-types/images/piplup.png
/kaggle/input/pokemon-images-and-types/images/trubbish.png
/kaggle/input/pokemon-images-and-types/images/dusknoir.png
/kaggle/input/pokemon-images-and-types/images/heatran.png
/kaggle/input/pokemon-images-and-types/images/ludicolo.png
/kaggle/input/pokemon-images-and-types/images/poliwhirl.png
/kaggle/input/pokemon-images-and-types/images/pansear.png
/kaggle/input/pokemon-images-and-types/images/gible.png
```

```
/kaggle/input/pokemon-images-and-types/images/lilligant.png
/kaggle/input/pokemon-images-and-types/images/morelull.png
/kaggle/input/pokemon-images-and-types/images/malamar.png
/kaggle/input/pokemon-images-and-types/images/meowth.png
/kaggle/input/pokemon-images-and-types/images/togekiss.png
/kaggle/input/pokemon-images-and-types/images/shedinja.png
/kaggle/input/pokemon-images-and-types/images/haxorus.png
/kaggle/input/pokemon-images-and-types/images/stunfisk.png
/kaggle/input/pokemon-images-and-types/images/nosepass.png
/kaggle/input/pokemon-images-and-types/images/shellder.png
/kaggle/input/pokemon-images-and-types/images/spritzee.png
/kaggle/input/pokemon-images-and-types/images/igglybuff.png
/kaggle/input/pokemon-images-and-types/images/chansey.png
/kaggle/input/pokemon-images-and-types/images/reshiram.png
/kaggle/input/pokemon-images-and-types/images/stoutland.png
/kaggle/input/pokemon-images-and-types/images/mothim.png
/kaggle/input/pokemon-images-and-types/images/muk.png
/kaggle/input/pokemon-images-and-types/images/musharna.png
/kaggle/input/pokemon-images-and-types/images/feebas.png
/kaggle/input/pokemon-images-and-types/images/celesteela.png
/kaggle/input/pokemon-images-and-types/images/chandelure.png
/kaggle/input/pokemon-images-and-types/images/uxie.png
/kaggle/input/pokemon-images-and-types/images/sawsbuck.png
/kaggle/input/pokemon-images-and-types/images/hippopotas.png
/kaggle/input/pokemon-images-and-types/images/moltres.png
/kaggle/input/pokemon-images-and-types/images/minior-meteor.png
/kaggle/input/pokemon-images-and-types/images/ribombee.png
/kaggle/input/pokemon-images-and-types/images/hoopa-confined.png
/kaggle/input/pokemon-images-and-types/images/wimpod.png
/kaggle/input/pokemon-images-and-types/images/cacnea.png
/kaggle/input/pokemon-images-and-types/images/pyroar.png
/kaggle/input/pokemon-images-and-types/images/magneton.png
/kaggle/input/pokemon-images-and-types/images/roggenrola.png
/kaggle/input/pokemon-images-and-types/images/monferno.png
/kaggle/input/pokemon-images-and-types/images/togedemaru.png
/kaggle/input/pokemon-images-and-types/images/clefable.png
/kaggle/input/pokemon-images-and-types/images/blitzle.png
/kaggle/input/pokemon-images-and-types/images/greninja.png
/kaggle/input/pokemon-images-and-types/images/exeggutor.png
/kaggle/input/pokemon-images-and-types/images/jynx.png
/kaggle/input/pokemon-images-and-types/images/drapion.png
/kaggle/input/pokemon-images-and-types/images/gabite.png
/kaggle/input/pokemon-images-and-types/images/omastar.png
/kaggle/input/pokemon-images-and-types/images/audino.png
/kaggle/input/pokemon-images-and-types/images/shieldon.png
/kaggle/input/pokemon-images-and-types/images/gigalith.png
/kaggle/input/pokemon-images-and-types/images/magearna.png
/kaggle/input/pokemon-images-and-types/images/ledyba.png
/kaggle/input/pokemon-images-and-types/images/riolu.png
```

```python
import pandas as pd
import numpy as np

df=pd.read_csv('/kaggle/input/pokemon-images-and-types/pokemon.csv')

df
```

```
            Name     Type1   Type2   Evolution
0      bulbasaur     Grass  Poison     ivysaur
1        ivysaur     Grass  Poison    venusaur
2       venusaur     Grass  Poison         NaN
3     charmander      Fire     NaN  charmeleon
4     charmeleon      Fire     NaN   charizard
..           ...       ...     ...         ...
804    stakataka      Rock   Steel         NaN
805  blacephalon      Fire   Ghost         NaN
806      zeraora  Electric     NaN         NaN
807       meltan     Steel     NaN         NaN
808     melmetal     Steel     NaN         NaN

[809 rows x 4 columns]
```

```python
df.isnull()
```

```
      Name  Type1  Type2  Evolution
0    False  False  False      False
1    False  False  False      False
2    False  False  False       True
3    False  False   True      False
4    False  False   True      False
..     ...    ...    ...        ...
804  False  False  False       True
805  False  False  False       True
806  False  False   True       True
807  False  False   True       True
808  False  False   True       True

[809 rows x 4 columns]
```

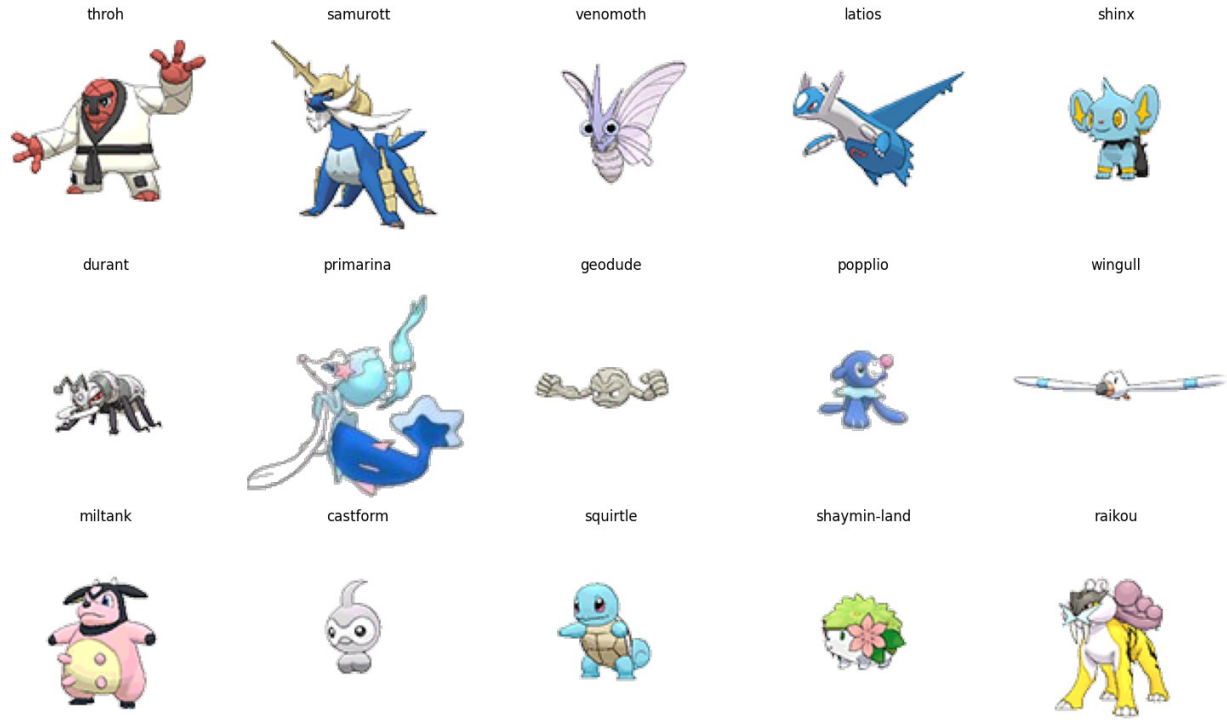
```python
df.isnull().sum()
```

```
Name           0
Type1          0
Type2        404
Evolution    777
dtype: int64
```

```python
print(os.listdir('../input'))
```

```
['pokemon-images-and-types']
```

```python
import os
import matplotlib.pyplot as plt
import tensorflow as tf

# Directory containing your images
image_folder = '/kaggle/input/pokemon-images-and-types/images'

# List all image file paths
image_paths = [os.path.join(image_folder, img) for img in
os.listdir(image_folder) if img.endswith(('png', 'jpg', 'jpeg'))]

# Limit to first 15 images (5 images per row for 3 rows)
image_paths = image_paths[:15]

# Display the images with their file names (without extensions)
plt.figure(figsize=(15, 9))  # Adjust the figure size to fit 15 images
nicely

for i, image_path in enumerate(image_paths):
    # Load the image
    image = tf.io.read_file(image_path)
    image = tf.image.decode_image(image)

    # Get the file name without the extension
    file_name = os.path.splitext(os.path.basename(image_path))[0]

    # Subplot for each image (3 rows, 5 columns)
    plt.subplot(3, 5, i + 1)
    plt.imshow(image)
    plt.title(file_name)  # Display the file name without extension as
the title
    plt.axis('off')  # No axes for a cleaner look

# Show all images
plt.tight_layout()
plt.show()
```




```
import os
import pandas as pd
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Flatten, Conv2D,
MaxPooling2D
from sklearn.model_selection import train_test_split
from tensorflow.keras.utils import to_categorical
import matplotlib.pyplot as plt

# Paths to dataset
data_folder = '/kaggle/input/pokemon-images-and-types'
images_folder = os.path.join(data_folder, 'images')
metadata_file = os.path.join(data_folder, 'pokemon.csv')

# Load metadata
metadata = pd.read_csv(metadata_file)

# Preprocess image data
def preprocess_image(image_path, label, img_size=(128, 128)):
    image = tf.io.read_file(image_path)
    image = tf.image.decode_image(image, channels=3)
    image = tf.image.resize(image, img_size)
    image = image / 255.0  # Normalize to [0, 1]
    return image, label
```

```python
# Convert labels to one-hot encoding
labels = metadata['Type1'].unique()
label_map = {label: idx for idx, label in enumerate(labels)}
metadata['Label'] = metadata['Type1'].map(label_map)
num_classes = len(labels)

# Prepare dataset
images = []
labels_list = []

for _, row in metadata.iterrows():
    image_path = os.path.join(images_folder, row['Name'] + '.png')  #
Ensure the file extension matches
    if os.path.exists(image_path):  # Check if the image file exists
        image, label = preprocess_image(image_path, row['Label'])
        images.append(image.numpy())  # Convert TensorFlow tensor to
NumPy array
        labels_list.append(label)

images = np.array(images)  # Convert list of arrays to a single NumPy
array
labels = np.array(labels_list)


# Split data into training and validation sets
X_train, X_val, y_train, y_val = train_test_split(images, labels,
test_size=0.2, random_state=42)


# Define the model
model = Sequential([
    Conv2D(32, (3, 3), activation='relu', input_shape=(128, 128, 3)),
    MaxPooling2D((2, 2)),
    Conv2D(64, (3, 3), activation='relu'),
    MaxPooling2D((2, 2)),
    Conv2D(128, (3, 3), activation='relu'),
    MaxPooling2D((2, 2)),
    Flatten(),
    Dense(128, activation='relu'),
    Dense(num_classes, activation='softmax')
])
```

```
/opt/conda/lib/python3.10/site-packages/keras/src/layers/
convolutional/base_conv.py:107: UserWarning: Do not pass an
`input_shape`/`input_dim` argument to a layer. When using Sequential
models, prefer using an `Input(shape)` object as the first layer in
the model instead.
  super().__init__(activity_regularizer=activity_regularizer,
**kwargs)
```

```python
# Compile the model
model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

# Train the model
history = model.fit(X_train, y_train, epochs=100,
validation_data=(X_val, y_val))
```

```
Epoch 1/100

WARNING: All log messages before absl::InitializeLog() is called are
written to STDERR
I0000 00:00:1726152430.454362      80 service.cc:145] XLA service
0x79f40c004b60 initialized for platform CUDA (this does not guarantee
that XLA will be used). Devices:
I0000 00:00:1726152430.454429      80 service.cc:153]   StreamExecutor
device (0): Tesla P100-PCIE-16GB, Compute Capability 6.0

17/21 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.1066 - loss:
2.8323

I0000 00:00:1726152433.389740      80 device_compiler.h:188] Compiled
cluster using XLA!  This line is logged at most once for the lifetime
of the process.

21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 75ms/step - accuracy: 0.1111 - loss:
2.8215

WARNING: All log messages before absl::InitializeLog() is called are
written to STDERR
I0000 00:00:1726152434.895196      81 asm_compiler.cc:369] ptxas
warning : Registers are spilled to local memory in function
'input_reduce_select_fusion_2', 64 bytes spill stores, 64 bytes spill
loads


21/21 ━━━━━━━━━━━━━━━━━━━━ 7s 124ms/step - accuracy: 0.1120 - loss:
2.8191 - val_accuracy: 0.1481 - val_loss: 2.7647
Epoch 2/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.2038 - loss:
2.5465 - val_accuracy: 0.1235 - val_loss: 2.8549
Epoch 3/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 0.2996 - loss:
2.2170 - val_accuracy: 0.1420 - val_loss: 2.9488
Epoch 4/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 0.4426 - loss:
1.7416 - val_accuracy: 0.1296 - val_loss: 3.5340
Epoch 5/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 0.6484 - loss:
1.0948 - val_accuracy: 0.1296 - val_loss: 4.0574
```

```
Epoch 6/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 0.8031 - loss:
0.6417 - val_accuracy: 0.0926 - val_loss: 5.1417
Epoch 7/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 0.8659 - loss:
0.3928 - val_accuracy: 0.0988 - val_loss: 5.9484
Epoch 8/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 0.9282 - loss:
0.2249 - val_accuracy: 0.0802 - val_loss: 6.3521
Epoch 9/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 0.9672 - loss:
0.1076 - val_accuracy: 0.0864 - val_loss: 7.6957
Epoch 10/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 0.9947 - loss:
0.0468 - val_accuracy: 0.0864 - val_loss: 9.2996
Epoch 11/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
0.0083 - val_accuracy: 0.1111 - val_loss: 9.5786
Epoch 12/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
0.0064 - val_accuracy: 0.1111 - val_loss: 10.1376
Epoch 13/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
0.0012 - val_accuracy: 0.1049 - val_loss: 10.6056
Epoch 14/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
0.0011 - val_accuracy: 0.1049 - val_loss: 10.8922
Epoch 15/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
6.2866e-04 - val_accuracy: 0.1049 - val_loss: 11.1038
Epoch 16/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
5.6029e-04 - val_accuracy: 0.1049 - val_loss: 11.3095
Epoch 17/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
4.6334e-04 - val_accuracy: 0.1049 - val_loss: 11.4881
Epoch 18/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
4.4022e-04 - val_accuracy: 0.1049 - val_loss: 11.6507
Epoch 19/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
4.1073e-04 - val_accuracy: 0.1049 - val_loss: 11.8054
Epoch 20/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
3.5785e-04 - val_accuracy: 0.1049 - val_loss: 11.9492
Epoch 21/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 1.0000 - loss:
3.3749e-04 - val_accuracy: 0.1049 - val_loss: 12.0782
Epoch 22/100
```

```
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
3.0557e-04 - val_accuracy: 0.0988 - val_loss: 12.2120
Epoch 23/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
2.6292e-04 - val_accuracy: 0.0988 - val_loss: 12.3302
Epoch 24/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
2.4365e-04 - val_accuracy: 0.0988 - val_loss: 12.4428
Epoch 25/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
2.1320e-04 - val_accuracy: 0.0988 - val_loss: 12.5524
Epoch 26/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
1.8391e-04 - val_accuracy: 0.0988 - val_loss: 12.6625
Epoch 27/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
1.9577e-04 - val_accuracy: 0.0988 - val_loss: 12.7618
Epoch 28/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
1.5994e-04 - val_accuracy: 0.0988 - val_loss: 12.8705
Epoch 29/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
1.8215e-04 - val_accuracy: 0.0988 - val_loss: 12.9696
Epoch 30/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
1.5793e-04 - val_accuracy: 0.0988 - val_loss: 13.0529
Epoch 31/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
1.4052e-04 - val_accuracy: 0.0988 - val_loss: 13.1433
Epoch 32/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
1.3026e-04 - val_accuracy: 0.0988 - val_loss: 13.2255
Epoch 33/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
1.1306e-04 - val_accuracy: 0.0988 - val_loss: 13.3065
Epoch 34/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
1.1032e-04 - val_accuracy: 0.0988 - val_loss: 13.3789
Epoch 35/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
1.1189e-04 - val_accuracy: 0.0988 - val_loss: 13.4575
Epoch 36/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
1.1794e-04 - val_accuracy: 0.0988 - val_loss: 13.5385
Epoch 37/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
9.0214e-05 - val_accuracy: 0.0988 - val_loss: 13.6069
Epoch 38/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
```

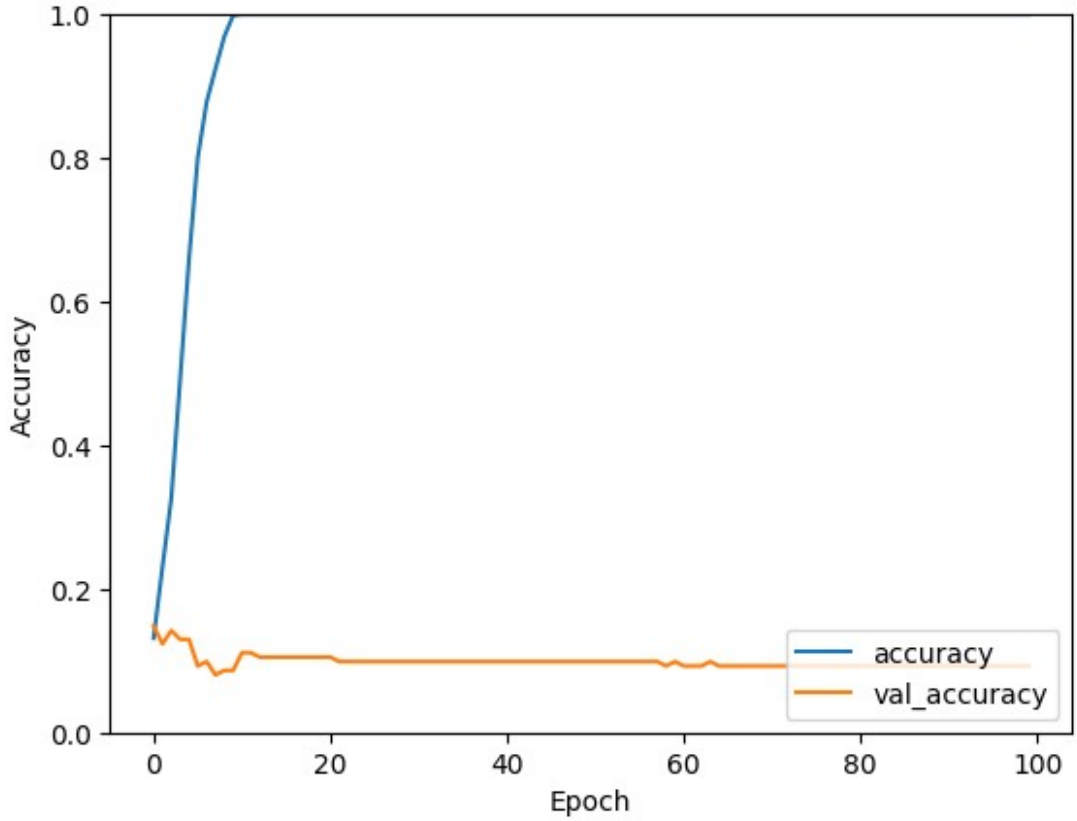
```
1.0061e-04 - val_accuracy: 0.0988 - val_loss: 13.6803
Epoch 39/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
8.5974e-05 - val_accuracy: 0.0988 - val_loss: 13.7455
Epoch 40/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
8.3077e-05 - val_accuracy: 0.0988 - val_loss: 13.8127
Epoch 41/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
7.7811e-05 - val_accuracy: 0.0988 - val_loss: 13.8772
Epoch 42/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
7.8511e-05 - val_accuracy: 0.0988 - val_loss: 13.9432
Epoch 43/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
7.7748e-05 - val_accuracy: 0.0988 - val_loss: 14.0038
Epoch 44/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
7.4425e-05 - val_accuracy: 0.0988 - val_loss: 14.0678
Epoch 45/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
7.8142e-05 - val_accuracy: 0.0988 - val_loss: 14.1315
Epoch 46/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
6.8106e-05 - val_accuracy: 0.0988 - val_loss: 14.1954
Epoch 47/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
7.1754e-05 - val_accuracy: 0.0988 - val_loss: 14.2512
Epoch 48/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
5.6091e-05 - val_accuracy: 0.0988 - val_loss: 14.3068
Epoch 49/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
5.8520e-05 - val_accuracy: 0.0988 - val_loss: 14.3592
Epoch 50/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
5.9217e-05 - val_accuracy: 0.0988 - val_loss: 14.4173
Epoch 51/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
4.8897e-05 - val_accuracy: 0.0988 - val_loss: 14.4759
Epoch 52/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
5.5759e-05 - val_accuracy: 0.0988 - val_loss: 14.5268
Epoch 53/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
5.0840e-05 - val_accuracy: 0.0988 - val_loss: 14.5852
Epoch 54/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
5.0829e-05 - val_accuracy: 0.0988 - val_loss: 14.6383
```

```
Epoch 55/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
5.2967e-05 - val_accuracy: 0.0988 - val_loss: 14.6868
Epoch 56/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
4.4963e-05 - val_accuracy: 0.0988 - val_loss: 14.7347
Epoch 57/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
3.7279e-05 - val_accuracy: 0.0988 - val_loss: 14.7798
Epoch 58/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
3.9169e-05 - val_accuracy: 0.0988 - val_loss: 14.8342
Epoch 59/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 1.0000 - loss:
3.8980e-05 - val_accuracy: 0.0926 - val_loss: 14.8903
Epoch 60/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
3.6737e-05 - val_accuracy: 0.0988 - val_loss: 14.9374
Epoch 61/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
3.5445e-05 - val_accuracy: 0.0926 - val_loss: 14.9848
Epoch 62/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
3.7008e-05 - val_accuracy: 0.0926 - val_loss: 15.0307
Epoch 63/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
3.5258e-05 - val_accuracy: 0.0926 - val_loss: 15.0788
Epoch 64/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
3.4371e-05 - val_accuracy: 0.0988 - val_loss: 15.1228
Epoch 65/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
2.8916e-05 - val_accuracy: 0.0926 - val_loss: 15.1673
Epoch 66/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
3.0482e-05 - val_accuracy: 0.0926 - val_loss: 15.2152
Epoch 67/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
3.3050e-05 - val_accuracy: 0.0926 - val_loss: 15.2563
Epoch 68/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
3.2665e-05 - val_accuracy: 0.0926 - val_loss: 15.3020
Epoch 69/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
2.8122e-05 - val_accuracy: 0.0926 - val_loss: 15.3407
Epoch 70/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
2.6645e-05 - val_accuracy: 0.0926 - val_loss: 15.3757
Epoch 71/100
```

```
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
2.7416e-05 - val_accuracy: 0.0926 - val_loss: 15.4212
Epoch 72/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
2.8647e-05 - val_accuracy: 0.0926 - val_loss: 15.4649
Epoch 73/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
2.6611e-05 - val_accuracy: 0.0926 - val_loss: 15.5046
Epoch 74/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
2.5844e-05 - val_accuracy: 0.0926 - val_loss: 15.5431
Epoch 75/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
2.5857e-05 - val_accuracy: 0.0926 - val_loss: 15.5816
Epoch 76/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
2.4248e-05 - val_accuracy: 0.0926 - val_loss: 15.6252
Epoch 77/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
2.6260e-05 - val_accuracy: 0.0926 - val_loss: 15.6651
Epoch 78/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
2.0982e-05 - val_accuracy: 0.0926 - val_loss: 15.7011
Epoch 79/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 1.0000 - loss:
2.0862e-05 - val_accuracy: 0.0926 - val_loss: 15.7367
Epoch 80/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
1.9710e-05 - val_accuracy: 0.0926 - val_loss: 15.7720
Epoch 81/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
2.2392e-05 - val_accuracy: 0.0926 - val_loss: 15.8116
Epoch 82/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
2.1778e-05 - val_accuracy: 0.0926 - val_loss: 15.8422
Epoch 83/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
1.7671e-05 - val_accuracy: 0.0926 - val_loss: 15.8803
Epoch 84/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
1.7515e-05 - val_accuracy: 0.0926 - val_loss: 15.9177
Epoch 85/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
1.5957e-05 - val_accuracy: 0.0926 - val_loss: 15.9535
Epoch 86/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
1.7409e-05 - val_accuracy: 0.0926 - val_loss: 15.9892
Epoch 87/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 1.0000 - loss:
```

```
1.6955e-05 - val_accuracy: 0.0926 - val_loss: 16.0287
Epoch 88/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 1.0000 - loss:
1.7310e-05 - val_accuracy: 0.0926 - val_loss: 16.0658
Epoch 89/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
1.7960e-05 - val_accuracy: 0.0926 - val_loss: 16.1017
Epoch 90/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
1.8242e-05 - val_accuracy: 0.0926 - val_loss: 16.1332
Epoch 91/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
1.6803e-05 - val_accuracy: 0.0926 - val_loss: 16.1697
Epoch 92/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
1.6415e-05 - val_accuracy: 0.0926 - val_loss: 16.2033
Epoch 93/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
1.5393e-05 - val_accuracy: 0.0926 - val_loss: 16.2378
Epoch 94/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
1.5419e-05 - val_accuracy: 0.0926 - val_loss: 16.2720
Epoch 95/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
1.4493e-05 - val_accuracy: 0.0926 - val_loss: 16.3038
Epoch 96/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
1.3202e-05 - val_accuracy: 0.0926 - val_loss: 16.3329
Epoch 97/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
1.4751e-05 - val_accuracy: 0.0926 - val_loss: 16.3673
Epoch 98/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
1.2928e-05 - val_accuracy: 0.0926 - val_loss: 16.4012
Epoch 99/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
1.4255e-05 - val_accuracy: 0.0926 - val_loss: 16.4355
Epoch 100/100
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 1.0000 - loss:
1.3066e-05 - val_accuracy: 0.0926 - val_loss: 16.4685
```

```python
# Plot training history (optional)
plt.plot(history.history['accuracy'], label='accuracy')
plt.plot(history.history['val_accuracy'], label='val_accuracy')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.ylim([0, 1])
plt.legend(loc='lower right')
plt.show()
```




```
# Save the model
model.save('/kaggle/working/model.h5')
```